



Biaxial tensile tests of the porcine ascending aorta



Valérie Deplano^{a,*}, Mourad Boufi^{a,b,c}, Olivier Boiron^a, Carine Guivier-Curien^a,
Yves Alimi^{b,c}, Eric Bertrand^a

^a Aix-Marseille Université, CNRS, Ecole Centrale Marseille, IRPHE UMR 7342, 13384 Marseille, France

^b Aix-Marseille Université, IFSTTAR, LBA UMR T24, 13916 Marseille, France

^c AP-HM, Service de Chirurgie Vasculaire de l'Hôpital Nord de Marseille, France

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ABSTRACT

One of the aims of this work is to develop an original custom built biaxial set-up to assess mechanical behavior of soft tissues. Stretch controlled biaxial tensile tests are performed and stereoscopic digital image correlation (SDIC) is implemented to measure the 3D components of the generated displacements. Using this experimental device, the main goal is to investigate the mechanical behavior of porcine ascending aorta in the more general context of human ascending aorta pathologies. The results highlight that (i) SDIC arrangement allows accurate assessment of displacements and so stress strain curves, (ii) porcine ascending aorta has a nearly linear and anisotropic mechanical behavior until 30% of strain, (iii) porcine ascending aorta is stiffer in the circumferential direction than in the longitudinal one, (iv) the material coefficient representing the interaction between the two loading directions is thickness dependent, (v) taking into account the variability of the samples the stress values are independent of the stretch rate in the range of values from 10^{-3} to 10^{-1} s^{-1} and finally, (vi) unlike other segments of the aorta, 4-month-old pigs ascending aorta is definitely not a relevant model to investigate the mechanical behavior of the human ascending aorta.

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1. Introduction

Numerous works deal with tensile testing of soft tissues such as arteries to characterize their mechanical responses. The objectives of these studies are various but whatever the goal and the used experimental method (for example uniaxial and/or biaxial tensile tests, bulge inflation test, inflation-extension test), the results depend on the conditions under which the tests are carried out. Therefore, tissue harvesting and storing, sample preparation, sample geometry, sample clamping method, sample thickness measurement, environmental conditions, accuracy of the dynamic acquisition, test protocols including preconditioning, initial loading, used range of tension or displacement, loading rate values, used metrology for displacement measurements, are of primary importance. According to Sacks and Sun (2003)'s work and the more recent reviews of Jacobs et al. (2013) and of Avanzini et al. (2014) an original custom built biaxial set-up has been thus developed, trying to take into account all the necessary precautions for successful mechanical characterization tests.

Using this new biaxial set-up the aim of this study was to discuss the relevance of porcine models to examine the biomechanics involved in type A dissections of the thoracic aorta. Whereas the prognosis of this disease has been improved by surgical treatment, the post-operative mortality remains high (from 15% to 30%, Ehrlich et al., 2000). Endovascular treatment could thus appear as an alternative but currently dedicated endoprosthesis does not exist for this particular aortic segment. With a long term goal to develop such endoprosthesis, the biomechanical behavior of healthy and pathological ascending thoracic aorta will be investigated. Porcine ascending aortas (PAsA) were so used to evaluate their capacity to firstly model healthy human ascending aorta. Only few experimental works using biaxial tensile tests have been carried out on PAsA: Nicosia et al. (2002), Gundiah et al. (2008), Martin et al. (2011), and Shah et al. (2014); the results obtained by the latter ones were in contrast to the others. The present study will thus determine the validity of the pig model by assessing the PAsA mechanical behavior. Finally, as controversial results still exist on the influence of the stretch rate value on the mechanical response (see Yang et al., 2011 for example), three different values of loading rate have been tested and the corresponding stress–strain curves have been compared.

* Corresponding author.

E-mail address: deplano@irphe.univ-mrs.fr (V. Deplano).

2. Methods

2.1. Tissue preparation

PAsa tubular structures (Fig. 1a) were harvested from fourteen 4-month-old pigs (mean weight 27.9 ± 3.6 kg), stored, refrigerated at 4°C in 0.9% NaCl solution and tested within 24 h after the tissue extraction. Experimental procedure was approved by the ethics board of the Surgical Center for Education and Research (CERC) at Marseille's North Hospital. Aortas were first cut between the sinotubular junction and the brachiocephalic trunk avoiding aorta elongation. The tubular structures of ascending aorta (Fig. 1a) were then longitudinally cut along the inner curvature (Fig. 1b). Two square samples of 20×20 mm, located on the anterior and posterior walls, were afterwards cut from each longitudinal cutting when their sizes permitted (Fig. 1c). A custom built cutting device was developed to obtain reproducible size samples. Finally, interstitial tissues were carefully removed from the adventitia side. 21 samples with a mean thickness, E_0 , of 2.18 ± 0.25 mm were finally tested. Special attention was paid on the thickness measurements. Pictures were taken of each lateral side of the undeformed samples and an image processing script using Matlab was developed to obtain the mean thickness value of each side. For each sample, a unique mean thickness value was then computed from the four sample lateral sides' mean thickness measurements.

2.2. Mechanical testing

Samples were mounted in our original home made biaxial set-up (Fig. 2a). The circumferential, θ , and longitudinal, L , directions of the sample were in line with the biaxial displacements. To attach sample to the displacement systems, 8 Kevlar suture lines with 16 barbless hooks regularly spaced and anchored on sample's sides were used (Figs. 2b and 3). A custom made template for positioning hooks on the sample was achieved to obtain uniformity of the strain fields within an area of 25% around the sample center (Fig. 4a and b) and thus to avoid edge effects (Eilaghi et al., 2009).

On each sample side, two pulleys were associated with two Kevlar suture lines to regulate potential difference in their length and ensure that the forces were equally distributed between each suture line. Shear-stresses were thus avoided or at least minimized leading to negligible shear strain values (Fig. 4c). Displacement-driven protocol was carried out while the ratio, α , remained constant. $\alpha = \frac{\lambda_\theta}{\lambda_L}$, noted $\alpha = \lambda_\theta : \lambda_L$, was defined as the stretch ratio, λ_θ and λ_L being the stretch in the θ , and L directions respectively.

The displacements were generated by 4 linear actuators (Parker) associated to 4 servomotors (Kollmorgen). The positions ($20 \mu\text{m}$ precision), velocities, and accelerations of each linear actuator were computer-controlled. Forces resulting from the displacements were measured using two load cells (10 ± 0.0015 N, 31E10, Honeywell) located on each direction of displacement. In addition to the recording of the imposed displacements, 3D displacement measurements were achieved using Stereoscopic Digital Image Correlation (SDIC). Stereoscopic arrangement protects from out of plane motion issues (Sutton et al., 2009). Two cameras (CMOS, 16-bit, 3Mpixels, 30 Hz) using 35 mm focal length were used. Speckle patterns on the intimal face of each sample (Fig. 3) were created using a fine tipped air-brush and Indian black ink (Genovese et al., 2011). Special attention was paid on the quality of the speckle patterns: density, size and repartition of gray levels and of the SDIC arrangement calibration. Samples were submerged in aqueous 0.9% NaCl solution maintained at 37°C during all the experiments. To optimize optical measurements a Plexiglass window was designed and positioned centrally over the sample in contact with the solution to avoid image deformations. A LED illumination was designed to ensure homogeneous lighting. The signals of forces and imposed displacements were acquired at 1 kHz. A specific trigger was generated at the start of each last cycle of the mechanical tests, to take, synchronized displacements, image pairs of the sample by the cameras. The acquisition frequency, f_a , of both cameras depended on the stretch rate values. Dantec Q-400 software was used to determine the 3D displacements (Fig. 5) and the deformation gradient F from each acquired pair of images at each grid point. The Green strain tensor of the

sample surface, E , was then inferred. Matlab scripts were developed to compute the mean values of E and λ (using the polar decomposition of F) within an area of 25% around the center of the sample for each acquired time and to manage recorded forces, displacements and data synchronizations. The accuracy of the functioning of our biaxial set-up components was periodically checked.

The maximum of the stretch, λ_{max} , was set to 1.3 of the gripped undeformed sample area i.e. $X_L \times X_\theta$, where $X_L = X_\theta = 18$ mm (Fig. 3) were measured before the start of the protocol. According to, Hansen et al. (1995) for example, 30% of strain was in the range of the physiological one. Twenty loading and unloading cycles of preconditioning were first performed at λ_{max} and $\alpha = 1 : 1$ following by $k=7$ consecutive cycles each one at different stretch ratios: 0.75:1, 1:0.75, 0.5:1, 1:0.5, 0.25:1, 1:0.25, 1:1. The last $\alpha = 1 : 1$ test was performed and compared to the last cycle of the preconditioning to check that no sample structural damage occurred during the overall protocol. Initial zero stress was assessed by tuning the position of the suture lines by step of 0.1 mm at very low displacement rate (0.1 mm/s) until the load cell signals detected were above their noise level (2.4 mV). At this time, forces were called initial forces, F_{init} . A hybrid displacement/force control was performed during the unloading path to prevent the sample being bended with respect to its original dimension. Through the monitoring of the forces as the sample was unloaded, the displacements were ceased in each direction if the forces were dropped to F_{init} .

This protocol was carried out for three different stretch rates $\dot{\lambda}_1 = 3.84 \times 10^{-3} \text{ s}^{-1}$, $\dot{\lambda}_2 = 3.84 \times 10^{-2} \text{ s}^{-1}$, and $\dot{\lambda}_3 = 3.47 \times 10^{-1} \text{ s}^{-1}$ on $NS=8$,

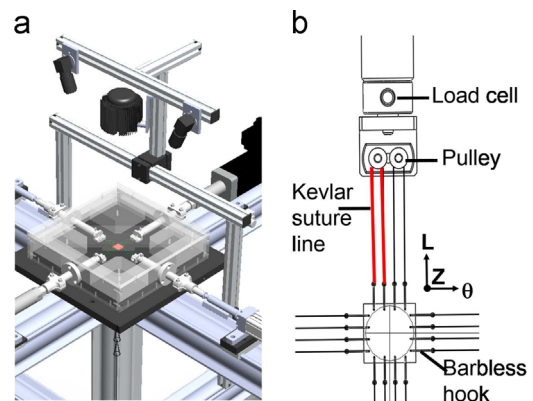


Fig. 2. (a) Experimental set-up. (b) Scheme of the top view.

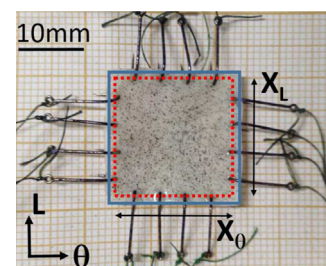


Fig. 3. Representative unloaded sample with its attachment system and its speckle patterns. The 20×20 mm blue square resulted from the square cut; the red dotted rectangle sizes 18×18 mm and represents the effective undeformed sample X_θ and X_L . The hooks' diameter is about 0.44 mm; the spacing between the hooks is about 4 mm. (For interpretation of the references to color in this figure caption, the reader is referred to the web version of this paper.)

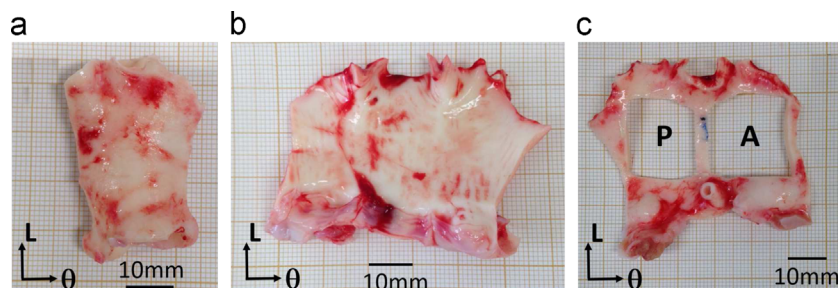


Fig. 1. (a) Representative sample of porcine ascending aorta. (b) Representative longitudinal cutting; view of the intimal face. (c) Representative of square cut; P for posterior sample and A for anterior one; view of the adventitia face.

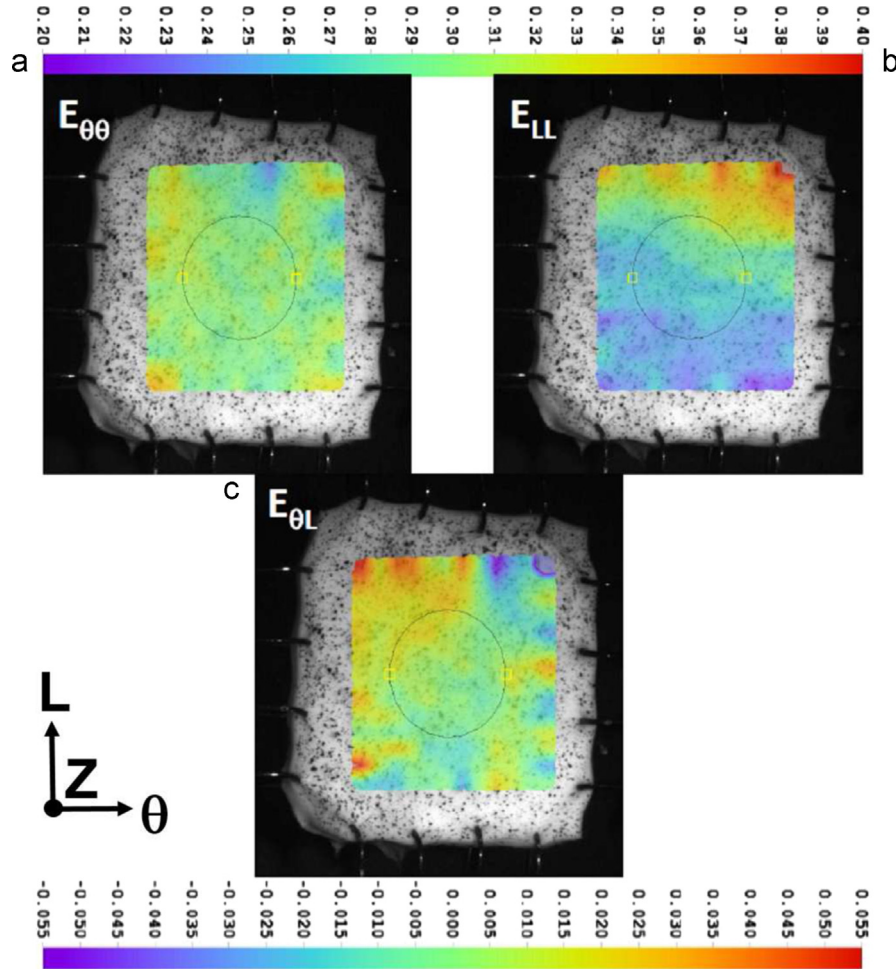


Fig. 4. Representative contours of (a) $E_{\theta\theta}$, (b) E_{LL} and (c) $E_{\theta L}$ obtained at the end of the loading path for $\alpha=1:1$, sample ID=6. Considering the area inside the black circle representing roughly 25% of the sample from its center, $\frac{E_{\theta\theta}}{E_{LL}}=1.11$ demonstrating that data were obtained from regions free of edges effects (O'Leary et al., 2014). $\bar{E}_{\theta L}=0.012\bar{E}_{\theta\theta}$, $\bar{E}_{\theta L}=0.014\bar{E}_{LL}$, demonstrating that shear strain was negligible.

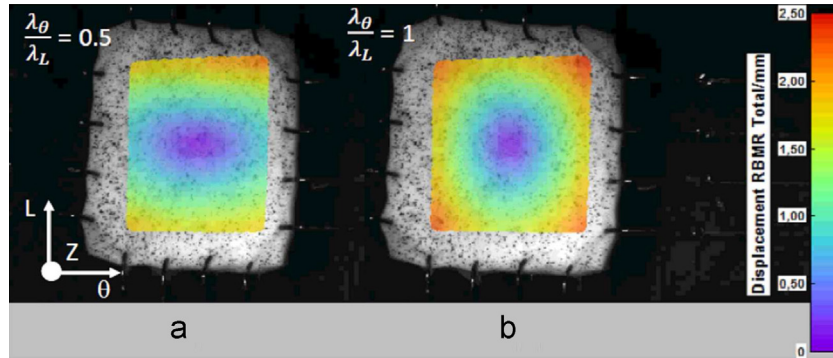


Fig. 5. Typical total displacement measurements obtained from SDIC for (a) $\alpha=0.5:1$ and (b) $\alpha=1:1$ at the end of loading path for sample ID=6.

NS=6, NS=7 different samples respectively. The λ_3 value was obtained from pig Peterson modulus, E_p (see Appendix A for details of measurements and calculus) using Bailly et al. (2014)'s work. The λ_1 value was selected as low value used in the literature (for example $\lambda=4 \times 10^{-3} \text{ s}^{-1}$ for Nicosia et al., 2002) and finally λ_2 was an intermediate value between the two others. Note that these stretch rates are the imposed and the maxima ones (those determined at λ_{max}). When the cameras were triggered, their maximum acquisition frequency was 15.15 Hz for a 16-bit pixel depth. The number of acquired image pairs for λ_3 was therefore limited to $2 \times n=26$ for one cycle using $f_a=15.15 \text{ Hz}$. For λ_1 and λ_2 , $2 \times n$ was equal to 118, using $f_a=0.756 \text{ Hz}$ and 7.57 Hz respectively.

The first Piola Kirchhoff stress tensor, P , is determined using f_θ and f_L , the recorded force, X_θ^{k-1} , X_L^{k-1} and E_0 , the undeformed gripped sample dimensions and thickness. For each cycle performed during one protocol, that is to say k consecutive cycles each one at different stretch ratios, X_θ^{k-1} , and X_L^{k-1} correspond to the undeformed lengths of the sample at the end of the $k-1$ unloading path; they could differ from the initial undeformed lengths X_θ and X_L in cases when the hybrid control is triggered. The components of P are expressed by $P_{\theta\theta}^k = \frac{f_\theta}{X_\theta^{k-1}E_0}$ and $P_{LL}^k = \frac{f_L}{X_L^{k-1}E_0}$. The components of the second Piola Kirchhoff stress tensor, S , are then obtained using $S_{\theta\theta}^k = \frac{P_{\theta\theta}^k}{\lambda_\theta}$ and $S_{LL}^k = \frac{P_{LL}^k}{\lambda_L}$. λ_i were calculated from the SDIC displacement measurements. For clarity, note that the superscript k will be omitted in the rest of the paper.

2.3. Statistical analysis

All statistical analyses were performed using statistical toolbox in Matlab. Unless otherwise specified, the comparison of the continuous variables in relation to the following different factors: pig weight, thickness sample, face sample, loading directions, and λ was performed using the unpaired two-sided Student t test or ANOVA when appropriate. These tests ensured that variables were normally distributed through Shapiro–Wilk test and variances were equals through F test or Bartlett test. All factors examined were subjected to univariate analysis. Statistical significance was defined as $p < 0.05$.

As proposed by Lin (1989) a concordance correlation coefficient $\hat{\rho}_c$ was used to determine the degree of agreement between two readings ($\hat{\rho}_c \in [0, 1]$, $\hat{\rho}_c = 1$ means a perfect agreement; see Annexe B for the mathematical definition of $\hat{\rho}_c$).

3. Results

All data analysis was performed for the loading part of the stress–strain curves and, unless otherwise specified, E and λ values are calculated from SDIC measurements.

3.1. Comparison of the two equibiaxial tests

The $S_{\theta\theta}$ and S_{LL} stress values of the first equibiaxial test (last preconditioning cycle) and the last one (last test of the protocol) were compared for same strain values using the concordance correlation coefficient, $\hat{\rho}_c \in [0.4233, 0.9999]$, with a mean value of 0.8996 and a median value of 0.9649. Therefore, as the concordance correlation coefficient is greater than 0.9 for 70% of all cases we concluded that a good concordance between the two equibiaxial tests was found, showing that the mechanical properties of the samples are not altered by the overall protocol.

3.2. General biaxial mechanical response

Fig. 6a and b shows representative S – E stress–strain curves for different values of α . The mechanical behavior of the PAsA is nearly linear whatever the direction. Anisotropic behavior is clearly observed. These results agree with those of Nicosia et al. (2002), Gundiah et al. (2008) and Martin et al. (2011) until strain values about 35%. It is worth noticing that Shah et al. (2014) results highlight anisotropic but non-linear behavior.

3.3. Constitutive modeling

Fig. 6 a and b shows an approximately linear relationship between stress and strain. According to these results, the mechanical response of PAsA is described by a 3-parameter polynomial strain energy function, W , as Nicosia et al. (2002) and Gundiah et al. (2008) did using the Vaishnav et al. (1972) model. W is given by $W = \frac{1}{2}(A_{\theta\theta}E_{\theta\theta}^2 + 2A_{\theta L}E_{\theta\theta}E_{LL} + A_{LL}E_{LL}^2)$, where $E_{\theta\theta}$ and E_{LL} are the components of the Green–Lagrangian strain tensor. The

components of the second Piola Kirchhoff stress tensor can be derived from W according to $S = \frac{\partial W}{\partial E}$. Applying this latter relation, linear stress–strain relationships are obtained: $S_{\theta\theta} = A_{\theta\theta}E_{\theta\theta} + A_{\theta L}E_{LL}$ and $S_{LL} = A_{\theta L}E_{\theta\theta} + A_{LL}E_{LL}$. The constants $A_{\theta\theta}$ and A_{LL} describe the tissue stiffness in the θ and L directions respectively and $A_{\theta L}$ is related to the interaction between the two directions. They were obtained using an optimization procedure (written in Matlab). For each PAsA sample, the mechanical data from the last preconditioning cycle and the 6 first tests of the biaxial protocol ($\alpha = 1:1$; 0.75:1; 1:0.75; 0.5:1; 1:0.5, 0.25:1; 1:0.25) were simultaneously fit to the constitutive model. The method was to minimize the sum of the error χ_α (Eq. (1)), e.g. $\sum \alpha \chi_\alpha$ where α was the stretch ratio and n was the number of the experimental data recorded during the loading path (i.e. $n = 59$ for $\lambda_1 = 3.84 \times 10^{-3} \text{ s}^{-1}$ and $\lambda_2 = 3.84 \times 10^{-2} \text{ s}^{-1}$ and $n = 13$ for $\lambda_3 = 3.47 \times 10^{-1} \text{ s}^{-1}$):

$$\chi_\alpha = \sum_{k=1}^n (S_{\theta\theta}^{\text{exp}} - S_{\theta\theta}^{\text{mod}})^2_k + (S_{LL}^{\text{exp}} - S_{LL}^{\text{mod}})^2_k \quad (1)$$

The superscript *exp* indicates the stress values from the experimental data, *mod* those predicted by the model.

Table 1 of Appendix C presents the A_{ij} coefficients obtained through the optimization procedure for the 21 samples obtained from 14 different pigs. Precautions were taken to ensure their uniqueness. The stress components derived from the strain energy function fit experimental data with average determination coefficients, R^2 , equals to 0.9538 and 0.9522 in the longitudinal and circumferential direction respectively. Fig. 6a and b highlights a typical fitting between experimental data and constitutive modeling.

3.4. Stiffness and anisotropy

Considering all the samples, $A_{\theta\theta} \in [157.173, 285.504] \text{ kPa}$ with $\bar{A}_{\theta\theta} = 212.332 \pm 39.901 \text{ kPa}$, and $\bar{A}_{\theta\theta} = 219.912 \text{ kPa}$, $A_{\theta L} \in [49.070, 105.319] \text{ kPa}$, with $\bar{A}_{\theta L} = 76.695 \pm 16.069 \text{ kPa}$, and $\bar{A}_{\theta L} = 75.912 \text{ kPa}$, $A_{LL} \in [118.895, 227.208] \text{ kPa}$, with $\bar{A}_{LL} = 174.887 \pm 32.439 \text{ kPa}$, and $\bar{A}_{LL} = 179.860 \text{ kPa}$. In addition, with a p value equal to 0.0018, $A_{\theta\theta}$ and A_{LL} were statistically different; PAsA was stiffer in the circumferential direction than in the longitudinal direction. These results agree with those of Nicosia et al. (2002), Gundiah et al. (2008) and Martin et al., 2011. The ratio $\beta = \frac{A_{\theta\theta} + A_{\theta L}}{A_{LL} + A_{\theta L}}$ can represent the anisotropy degree of the tissue. $\beta \in [0.835, 1.476]$ with $\bar{\beta} = 1.163 \pm 0.178$ and $\bar{\beta} = 1.154$.

No statistically significant difference was found in the stiffness between the anterior and the posterior samples: $p = 0.09$ for $A_{\theta\theta}$, $p = 0.77$ for A_{LL} . The median value of weight (respectively thickness) was chosen as a threshold to form 2 groups. No statistically significant difference was found in the stiffness (i) in the weight groups: $p = 0.07$ for $A_{\theta\theta}$, $p = 0.39$ for A_{LL} and (ii) in the thickness

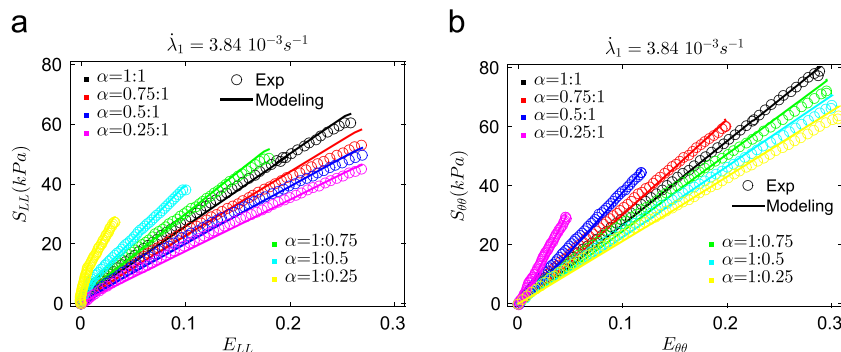


Fig. 6. Illustrative S – E curves plotted at $\lambda_1 = 3.84 \times 10^{-3} \text{ s}^{-1}$ for all α ratios; sample ID=9; symbols for experimental data, solid for constitutive modeling. (a) L direction. (b) θ direction.

groups: $p=0.23$ for A_{LL} , $p=0.34$ for $A_{\theta\theta}$ (for this latter case a Mann–Whitney U test was done because normality assumption was not obtained for one group).

The independence of the tissue stiffness to the anterior posterior samples disagrees with the results of Nicosia et al. (2002) but is in line with the results of Gundiah et al. (2008).

Whereas no statistically significant difference was found in $A_{\theta L}$ between anterior and posterior samples ($p=0.15$), and for weight ($p=0.14$), it is important to underline that significant difference was found for the thickness with a p value of 0.013. When the sample thickness decreases $A_{\theta L}$ increases.

3.5. Influence of $\dot{\lambda}$

Fig. 7a and b presents the S – E curves behavior for $\alpha = 0.75 : 1$ and for the three imposed values of $\dot{\lambda}$. For each group of $\dot{\lambda}$, the plotted stress values were the mean ones $\pm$ the standard deviations. These curves show that, considering the sample variability within a $\dot{\lambda}$ group, actually represented by the standard deviations, the stress values are independent of the loading rate. This tendency observed for $\alpha = 0.75 : 1$ can be generalized for all the stretch ratios; actually no statistically significant difference was

found in A_{ij} coefficients between the 3 different $\dot{\lambda}$ values; $p=0.13$ for $A_{\theta\theta}$, $p=0.23$ for $A_{\theta L}$, and $p=0.24$ for A_{LL} .

3.6. SDIC versus imposed λ measurements

Fig. 8 shows the P – λ curves plotted either using the λ values from the SDIC displacement measurements or the motor displacement measurements (mechanically imposed λ), for the different values of α for sample ID=7. On average, for a given value of P stress, the imposed stretch is over estimated by only 3.2% and 2.4% in the longitudinal and circumferential direction respectively. However, for a fixed value of stretch, imposed one induces a mean under estimation of the stress values by 45.4% and 21.05% in the longitudinal and circumferential direction respectively.

4. Discussion and conclusion

A new custom built biaxial set-up has been developed to assess mechanical behavior of soft tissues. Through stretch controlled biaxial tensile tests using this set-up, the present work has put into light that (i) SDIC arrangement allows accurate assessment of displacements, (ii) porcine ascending aorta of 4 month old pigs has

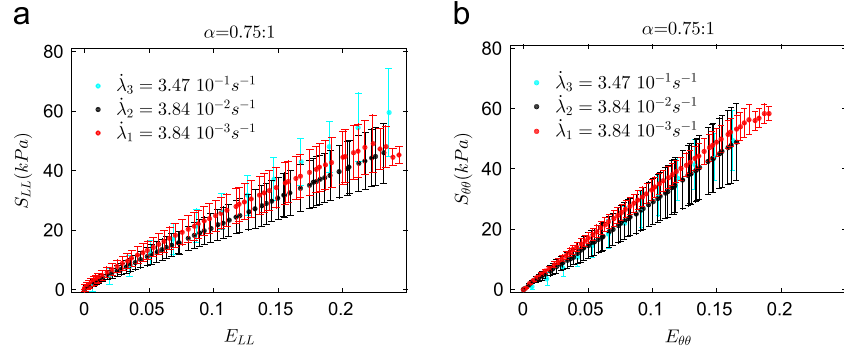


Fig. 7. S – E curves plotted at $\alpha=0.75:1$ for all $\dot{\lambda}$ values. (a) L direction. (b) θ direction. Data were presented as a mean $\pm$ standard deviation.

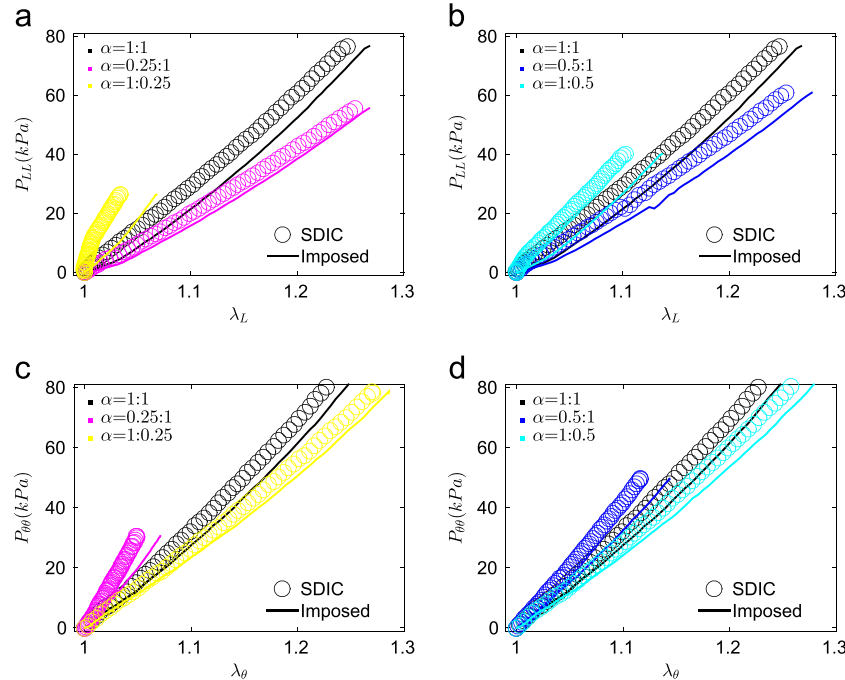


Fig. 8. P – λ curves for sample ID=7 at different values of α . Symbols for SDIC measurements, solid lines for imposed ones. P_{LL} – λ_L for (a) $\alpha = 1:1$, $\alpha = 0.25:1$, $\alpha = 1:0.25$ and (b) $\alpha = 1:1$, $\alpha = 0.5:1$, $\alpha = 1:0.5$. $P_{\theta\theta}$ – λ_θ for (c) $\alpha = 1:1$, $\alpha = 0.25:1$, $\alpha = 1:0.25$ and (d) $\alpha = 1:1$, $\alpha = 0.5:1$, $\alpha = 1:0.5$. $\alpha = 1:1$ corresponds to last test of the protocol.

a nearly linear and anisotropic mechanical behavior until 30% of strain, (iii) PAsA is stiffer in the circumferential direction than in the longitudinal one, (iv) the material coefficient representing the interaction between the two loading directions is thickness dependent, (v) taking into account the variability of the samples the stress values are independent of the stretch rate in the range of values from 10^{-3} to 10^{-1} s^{-1} and finally, (vi) considering 30% of deformation, unlike other segments of the aorta, 4-month-old pig ascending aorta, is definitely not a relevant model to investigate the mechanical behavior of the human ascending aorta.

Nicosia et al. (2002), Gundiah et al. (2008), Martin et al. (2011), and Shah et al. (2014), who worked on porcine ascending aorta mechanical characterization using biaxial tensile tests, used different sample parameters as well as varied testing protocols and measurement procedures (Table 1). Therefore, care should be taken when comparing the mechanical behavior they obtained with our results. In agreement with our work, Nicosia et al. (2002), Gundiah et al. (2008), and Martin et al. (2011), highlighted that the anisotropic PAsA mechanical behavior was linear until strain values about 30%. Nevertheless, Nicosia et al. (2002) found circumferential and longitudinal stiffness values ($\bar{A}_{\theta\theta} = 169.9 \pm 7.36 \text{ kPa}$ and $\bar{A}_{LL} = 115.8 \pm 8.39 \text{ kPa}$) lower than ours ($\bar{A}_{\theta\theta} = 212.332 \pm 39.901 \text{ kPa}$ and $\bar{A}_{LL} = 174.887 \pm 32.439 \text{ kPa}$). However, the dimension of their undeformed sample was not measured between the grips, the adventitia of their aorta samples was dissected away and this layer is considered to be less extensible compared to the media that contains a much greater fraction of type III collagen (Zeinali-Davarani et al., 2015). We used grip to grip lengths for undeformed sample dimension and adventitia was kept intact in our experiment so the stiffness discrepancy could be explained. However, it would be important to check that the pH value equal to 6 of the standard 0.9% saline solution used in the present work does not rigidify the tissue samples. Shah et al. (2014) observed a non-linear mechanical behavior of PAsA but their stress strain curves seem to be linear until 25% of deformation and the stress strain curves were plotted using grip strain. To conclude with the comparisons, the key point is that all these authors observed a nearly linear and anisotropic mechanical behavior until 25%.

Among several parameters, it is well known that the stiffness of healthy aortic wall is related to the connective fibers (elastin and collagen) (Tsamis et al., 2013). It was found in the present study that there was no significant difference in the tissue stiffness with the thickness E_0 ; however there was one with the coupling coefficient $A_{\theta L}$. Therefore one can hypothesize that the tissue stiffness depends on the fibre quantity rather than their apparent density. Consequently, for a same quantity of fibers if the wall thickness decreases their interaction increases. This phenomenon seems linked with the significant difference found in $A_{\theta L}$ material coefficient with thickness.

It has been found that arterial walls exhibit hysteresis that is relatively insensitive to strain rate over several decades (Fung, 1993; Holzapfel et al., 2002). Okamoto et al. (2002) observed that equibiaxial stress–strain curves obtained for human ascending aorta were similar when stretching at strain rates between $0.5\% \text{ s}^{-1}$ and $6\% \text{ s}^{-1}$. Yang et al. (2011) have more recently put into evidence that stress is insensitive to strain rate in arterial tissue when the loading rate is low. Their experimental tests were achieved on porcine abdominal arteries and strain rates from 0.0049 length/s to 0.29 length/s . They reported that controversial results found on this subject (Mohan and Melvin, 1982; Stemper et al., 2007) versus those in agreement (Tanaka and Fung, 1974; Lee and Haut, 1992) mainly depend on the control of the testing environment. The present work leads to confirm that the stress values are insensitive to deformation rates varying from 0.38 to $34.8\% \text{ s}^{-1}$. The minimum and maximum λ values, chosen in this

Table 1

Summary of test procedures performed on PAsA using biaxial tensile tests. Abbreviations: NPC: Number of PreConditioning cycles. SH: Slaughter House. AC: Animal Center. A: Age of pig in months. TT: maximum elapsed Time between harvesting and Test. W/OAd: sample WithOut Adventitia. NS: Number of Samples. W/OAd: sample WithOut Adventitia and Intima. DUS: Dimension of Unloaded Sample. AM: Attachment method. DUGtG: Grip to Grip Dimension of Unloaded sample. *: Details not provided. RT: Room Temperature. RaT: Ratio of Tension. DC: Displacement Controlled. SC: Stress Controlled. OPT: 4 or 5 graphic or ceramic markers and 1 CCD camera. fa: camera acquisition frequency. DIC: Digital Image Correlation with speckle patterns and 1 CCD camera. SDIC: Stereo Digital Image Correlation with speckle patterns and 2 CMOS cameras.

Authors	Sample	Bath	Preconditioning	Protocol	Strain	Loading rate	Strain measurement
Nicosia et al. (2002)	SH: A=*; TT=72 h; W/OAd: NS=36; DUS:20 × 20 mm; AM:suture; DUGtG:*	Neutral buffered medium 37 °C	Yes but *	DC: Preload:*; α=1:1; 1.7:1; 1.35:1; 1:1.35; 1:1.7	4 mm for 1:1	0.08 mm s ⁻¹	OPT, fa=15 Hz
Gundiah et al. (2008)	SH: A=*; TT=24 h; NS=10; DUS:23.2 × 23.2 mm; AM:suture; DUGtG:*	Saline, RT	NPC:10 for α=1:1 to 10% stretch	DC: Preload:0.N; α=1:1	10, 15, 20, 35%	4.06 mm s ⁻¹	OPT, fa=30 Hz
Martin et al. (2011)	AC: A=6–9; TT=–80 °C freezer; NS=10; DUS:21.5 × 21.5 mm; AM:suture; DUGtG:*	0.9%NaCl; RT	NPC:20	SC: Preload:*; RaT=0.75:1; 0.5:1; 0.3:1; 1:1; 1:0.75; 1:0.5; 1:0.3	At the maximum load without causing damage	*	OPT, fa=*
Shah et al. (2014)	SH: A=6; TT=24 h; W/OAd: NS=9; DUS: cruciform; AM:clamp; DUGtG:15 × 15 mm	1% PBS, RT	NPC:9 for α=1:1 to 40% of grip strain	DC: Preload:0.1N; α=1:1	40% of grip strain	0.05 mm s ⁻¹	DIC, fa=0.2 Hz
Present work	AC: A=3–6; TT=24 h; NS=21; DUS:20 × 20 mm; AM:suture; DUGtG:18 × 18 mm	0.9%NaCl, 37 °C	NPC:20 for α=1:1 to 30% of DUGtG	DC: Preload:0.N; α=1:1; 0.75:1; 1:0.75; 0.5:1; 1:0.5; 0.25:1; 1:0.25; 1:1	30% of DUGtG	3.13 mm s ⁻¹ 0.34 mm s ⁻¹ 0.034 mm s ⁻¹	SDIC, fa=15.15 Hz, 7.56 Hz, 0.756 Hz

study, are representative of characteristic values used in the literature and of pig physiological stretch rate obtained through in vivo measurements respectively. Consequently, one of the issues mentioned in the literature (Duprey et al., 2010; Shah et al., 2014), including works of our group (Bailly et al., 2014), to assess mechanical properties of tissues using non-physiological strain rate seems to be unfounded at least with deformation rate values investigated in this study, preconditioning, controlled testing environment and considering the variability of the tested samples. One limitation of the present work concerns the existence of residual stresses. They were neither measured nor taken into account in the constitutive equations of the model considering that the samples were in a stress-free state. Measurements of the opening angle and the longitudinal retraction upon aortic tubular segment excision are on going.

Unlike PAsA, human ascending aorta mechanical behavior is non-linear (Martin et al., 2011; Roccabianca et al., 2013 for example). This work is therefore on going with biaxial tensile tests carry on healthy human ascending aortas from organ donors and, on pathological ones from patients with type A aortic dissections undergoing surgical treatment. Histological analysis as well as microstructure one using multiphoton microscopy are also performed to link aortic wall constituents at the microscopic scale with the mechanical tissue characterizations.

Conflict of interest statement

All authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data associated with this paper can be found in the online version at <http://dx.doi.org/10.1016/j.jbiomech.2016.05.005>.

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